

RegimeForgeEA: Detailed Research and Test Specification for the Adaptive M5 Volume-Confirmed Reversal Candidate

RegimeForgeEA Research

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Contents

Executive statement	2
1. Objective and decision boundary	2
2. Data provenance and scope	2
2.1 Proxy data	2
2.2 Cost model	2
2.3 Position-size convention	2
3. Base alpha factor	2
4. Execution model	3
5. Candidate selection protocol	3
5.1 Candidate universe	3
5.2 Training gate	3
5.3 Validation and holdout gates	3
6. Base-rule results	4
7. Drawdown investigation	4
7.1 Observed concentration	4
7.2 Why position size alone is insufficient	4
7.3 Why the overlay is exploratory	4
8. Adaptive risk overlay	4
8.1 Completed-H1 trend confirmation	4
8.2 Daily realized-loss lock	4
8.3 Consecutive-loss cooldown	5
9. Adaptive backtest results	5
10. MQL5 implementation specification	5
11. MT5 Strategy Tester protocol	7
12. Failure modes and rejection criteria	7
13. Reproducibility commands	7
14. Final conclusion	8

Executive statement

This document is the detailed technical, quantitative, and operational record for the current M5 volume-confirmed reversal candidate. It is intentionally more conservative than a marketing report. It distinguishes between evidence, assumptions, exploratory changes, and unproven claims.

The base factor passed a pre-defined proxy-data sequence: 2021-2023 training, 2024 validation, and a single 2025 holdout. The adaptive overlay was created after reviewing the base rule's 2025 drawdown. Therefore, the adaptive result is an exploratory risk-control result, not an untouched out-of-sample result. It must be retested on independent broker-native XAUUSD Bid/Ask data before it is considered suitable for any demo or live deployment.

1. Objective and decision boundary

The objective is not to prove that the strategy will make money. No historical backtest can establish that. The objective is to identify a rule with a repeatable proxy-data edge, make every execution assumption explicit, and provide an MT5 implementation that a professional tester can invalidate or confirm with broker-native data.

The strategy is permitted to open long positions only. It is designed to buy a short-term oversold move when the bar's activity is unusually high, while the completed H1 trend remains positive. This is a conditional mean-reversion strategy, not a trend-following or grid system. It never averages down, never adds to an open position, and permits at most one managed position at a time.

2. Data provenance and scope

2.1 Proxy data

Research data consists of weekday-only PAXGUSD M5 bars from 2021 through 2025. PAXG is a gold-backed token and PAXGUSD is used as a public, gold-linked proxy. The associated trade volume is derived from public aggregate trade archives. Archive checksums were verified during acquisition.

This source offers a long, reproducible public history. It does **not** provide the exact liquidity, spread, session calendar, execution rules, tick volume, or contract economics of any XAUUSD broker. Every proxy result in this document must be read with that limitation.

2.2 Cost model

The order-level model applies a fixed 0.35 USD round-trip spread. A long entry uses next-bar open plus half-spread. The fixed-time exit uses the configured exit-bar close minus half-spread. No commission, swap, slippage, delay, partial fill, margin rejection, trading halt, or spread expansion is modeled.

This cost model is deliberately simple. It is neither a best-case XAUUSD model nor a substitute for MT5 real-tick testing. A professional tester must replace it with actual bid/ask, commission, swap, and slippage behavior.

2.3 Position-size convention

Backtest dollar PnL uses a 0.10 lot / 100 oz reference only to make PnL and drawdown intuitive. Profit factor, trade win rate, and net basis points do not depend on this reference under the stated fixed-cost model. The current MQL5 EA defaults to 0.02 lot as a conservative test volume, not as a universal sizing recommendation.

3. Base alpha factor

Let C_t be the close of the completed M5 signal bar and V_t its volume. The short-horizon return is:

$$r_3(t) = \frac{C_t}{C_{t-3}} - 1$$

The lower threshold is a trailing 20th percentile over 20 trading days, or 5,760 M5 bars:

$$q_{20}(t) = Q_{0.20}(\{r_3(s) : s < t\})$$

The implementation shifts the threshold one completed bar. Consequently, the signal bar is not included in its own reference distribution. This prevents a future-data or self-inclusion error.

The volume surprise ratio is:

$$VR(t) = \frac{V_t}{SMA_{60}(V)_t}$$

The base entry condition is:

$$r_3(t) \leq q_{20}(t) \quad \wedge \quad VR(t) \geq 1.5$$

Interpretation: a three-bar downside move alone is not sufficient. The move must occur with volume at least 50% above the trailing 60-bar average. The rule attempts to capture an oversold liquidation or exhaustion event rather than a routine price fluctuation.

4. Execution model

The chronological sequence for a signal is as follows:

1. M5 bar t closes.
2. The strategy calculates $r_3(t)$, the shifted quantile, and $VR(t)$.
3. If all entry filters pass, it enters on the first tradable price of bar $t + 1$.
4. The position remains open for 24 M5 bars.
5. It exits after the close of bar $t + 24$.
6. No second entry is allowed while the first position is open.

The fixed 24-bar horizon equals 120 minutes. This horizon was one of the pre-defined candidate choices in the order-level grid. It is not adjusted by trade outcome, and the rule contains no martingale, recovery sizing, or unbounded averaging mechanism.

5. Candidate selection protocol

5.1 Candidate universe

The direct M5 grid examined order-flow momentum, order-flow reversal, order-flow absorption, return reversal, breakout reversal, and volume-confirmed return reversal. The grid considered long and short signals, all-hours and regional UTC sessions, and 5, 10, 15, 30, 60, and 120 minute holding horizons.

5.2 Training gate

The 2021-2023 training gate required all of the following:

- At least 100 executed, non-overlapping trades.
- Profit factor of at least 1.10.
- Mean net return of at least 0.50 basis points per trade.

Candidates passing the gate were ranked by mean net basis points multiplied by the square root of trade count. This score favors both average edge and sample size, while avoiding a direct optimization of the 2025 holdout.

5.3 Validation and holdout gates

The training-selected row had to show at least 30 trades, PF of at least 1.05, and mean net return of at least 0.25 bps in 2024. Only after validation passed was the chosen base rule run once on 2025.

The selected base rule passed all three proxy periods. This is useful evidence, but it is not a guarantee, because PAXGUSD is not XAUUSD and the trading-cost model is incomplete.

6. Base-rule results

Sample	Trades	Net PnL (USD)	PF	Win rate	Mean net bps	Maximum drawdown (USD)
Training, 2021-2023	4,333	11,644.50	1.156	51.63%	1.512	-2,111.00
Validation, 2024	1,428	7,832.00	1.257	55.04%	2.447	-1,847.00
Holdout, 2025	1,350	6,707.20	1.119	55.41%	1.768	-4,927.40

The base rule has positive PF in each sample. Its critical weakness is not the average trade; it is the tail behavior of a long-only reversion strategy during persistent adverse regimes. The 2025 maximum drawdown is large relative to the reference capital of a typical small test account.

7. Drawdown investigation

7.1 Observed concentration

The deepest base-rule 2025 drawdown occurred on 21 November 2025. The worst 2025 monthly PnL observations occurred in May, October, and December. This pattern is consistent with a reversion rule repeatedly entering while downside pressure remains persistent, rather than with a uniform deterioration across all market conditions.

7.2 Why position size alone is insufficient

Lowering lots scales nominal drawdown but does not correct a regime mismatch. For example, a 0.02 lot test volume reduces the approximate dollar exposure by 80%, but the strategy can still experience clustered losing signals. Position sizing is necessary for account survivability; it is not a replacement for a market-state filter or a risk lock.

7.3 Why the overlay is exploratory

The adaptive controls were proposed after the base-rule 2025 drawdown had been observed. The result is useful for risk engineering, but it is no longer a purely untouched holdout test. A new independent period, or broker-native XAUUSD history unavailable during design, is required to assess the overlay without this information leak.

8. Adaptive risk overlay

The overlay leaves the alpha formula unchanged. It only determines whether a base signal may open a new position.

8.1 Completed-H1 trend confirmation

Let $EMA_{20}^{H1}(t-1)$ and $EMA_{50}^{H1}(t-1)$ be values on the most recently completed H1 bar. Permit a long M5 signal only if:

$$EMA_{20}^{H1}(t-1) > EMA_{50}^{H1}(t-1)$$

The H1 shift is important. A strategy that uses the unfinished current H1 bar can leak future intrabar information in a bar-based test.

8.2 Daily realized-loss lock

Let B_d be the account balance at the broker-day start and R_d the realized PnL for the current day. Block future entries once:

$$R_d \leq -0.02B_d$$

The EA uses balance, not floating equity, for this lock. It does not force-close the open trade. It simply avoids compounding realized losses with new entries until the next broker day.

8.3 Consecutive-loss cooldown

After four consecutive losing exits, the EA opens no new trade for 12 M5 bars. The loss counter resets after a profitable exit. This is a simple, observable state machine intended to reduce entry clustering during a failed reversion episode.

9. Adaptive backtest results

Sample	Trades	Net PnL (USD)	PF	Win rate	Mean net bps	Maximum drawdown (USD)
Training, 2021-2023	2,029	6,063.90	1.162	52.34%	1.714	-2,251.00
Validation, 2024	790	5,565.00	1.334	56.58%	3.131	-946.00
Holdout, 2025	828	7,151.00	1.221	56.64%	2.768	-3,134.50

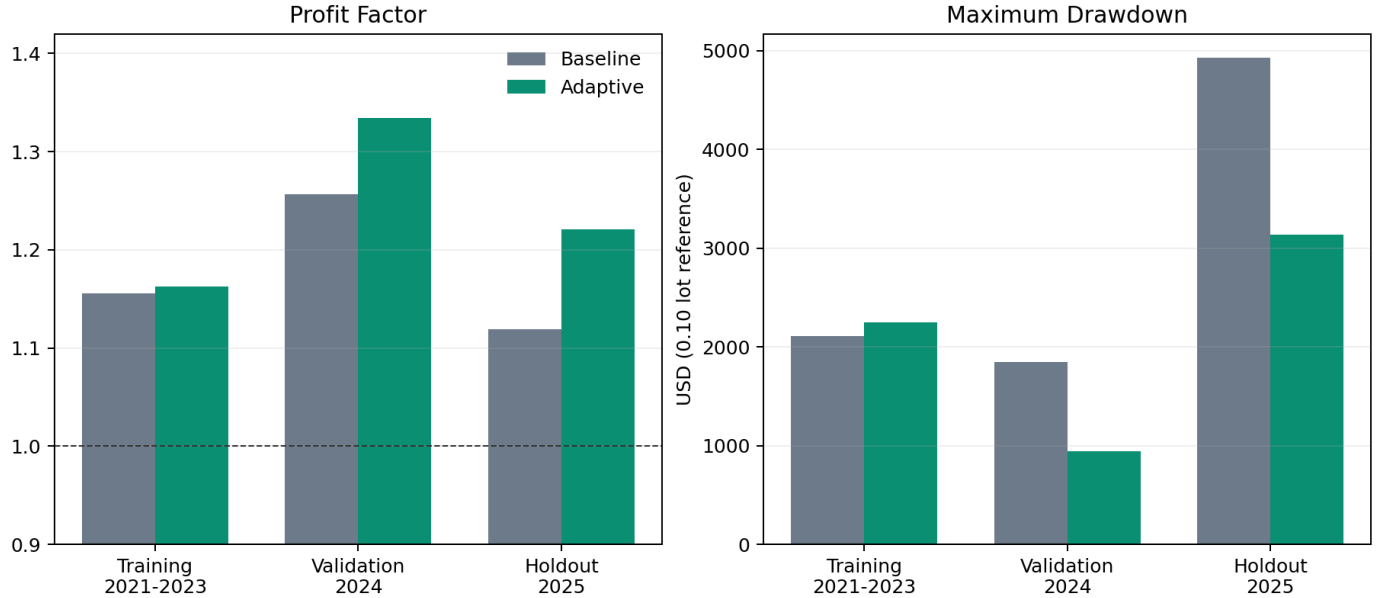


Figure 1: Base and adaptive PF/drawdown comparison

The overlay improves PF in all displayed periods. It improves maximum drawdown substantially in 2024 and 2025 but makes the training drawdown slightly worse. The correct interpretation is not that the overlay is globally optimal. It is a reasonable risk-control hypothesis that requires a fresh out-of-sample test.

10. MQL5 implementation specification

The implementation is located at `Experts/RegimeForgeVolumeReversalEA.mq5`. Its operational behavior is designed to match the Python research semantics as closely as an event-driven MT5 EA can.

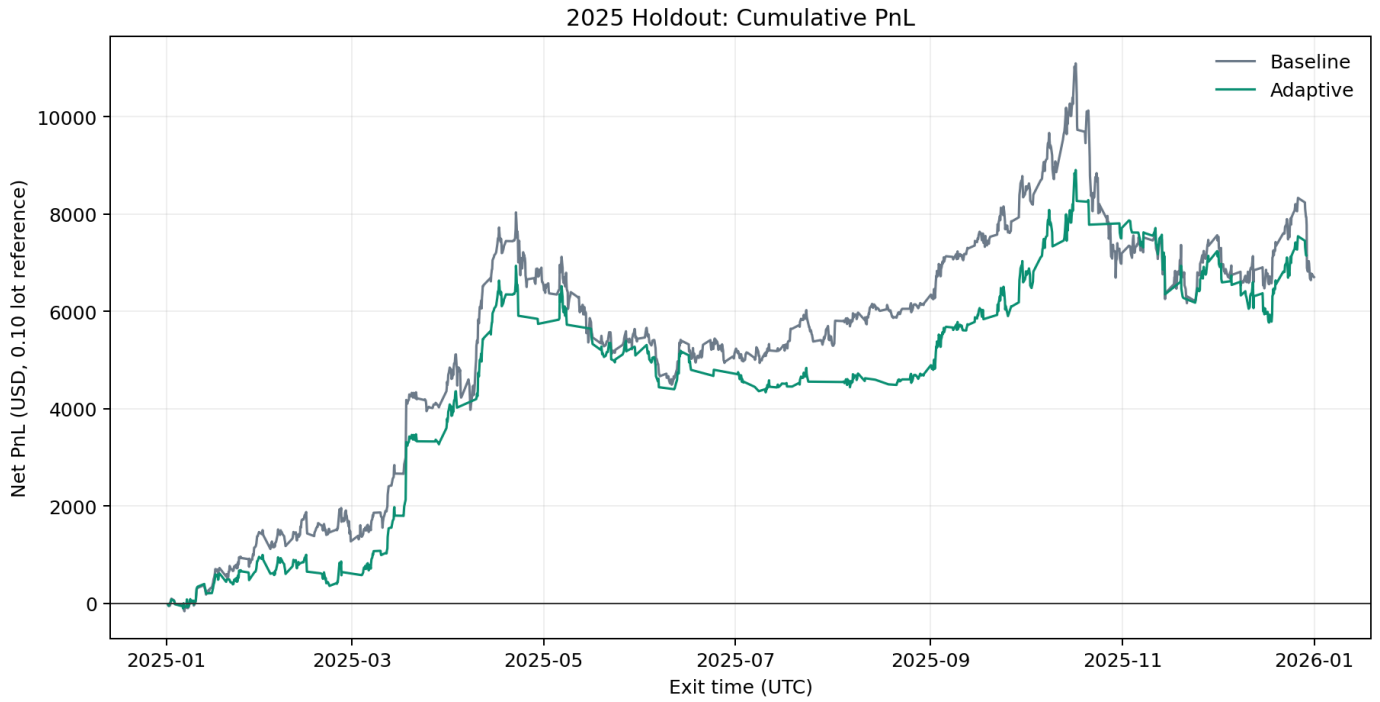


Figure 2: 2025 holdout equity comparison

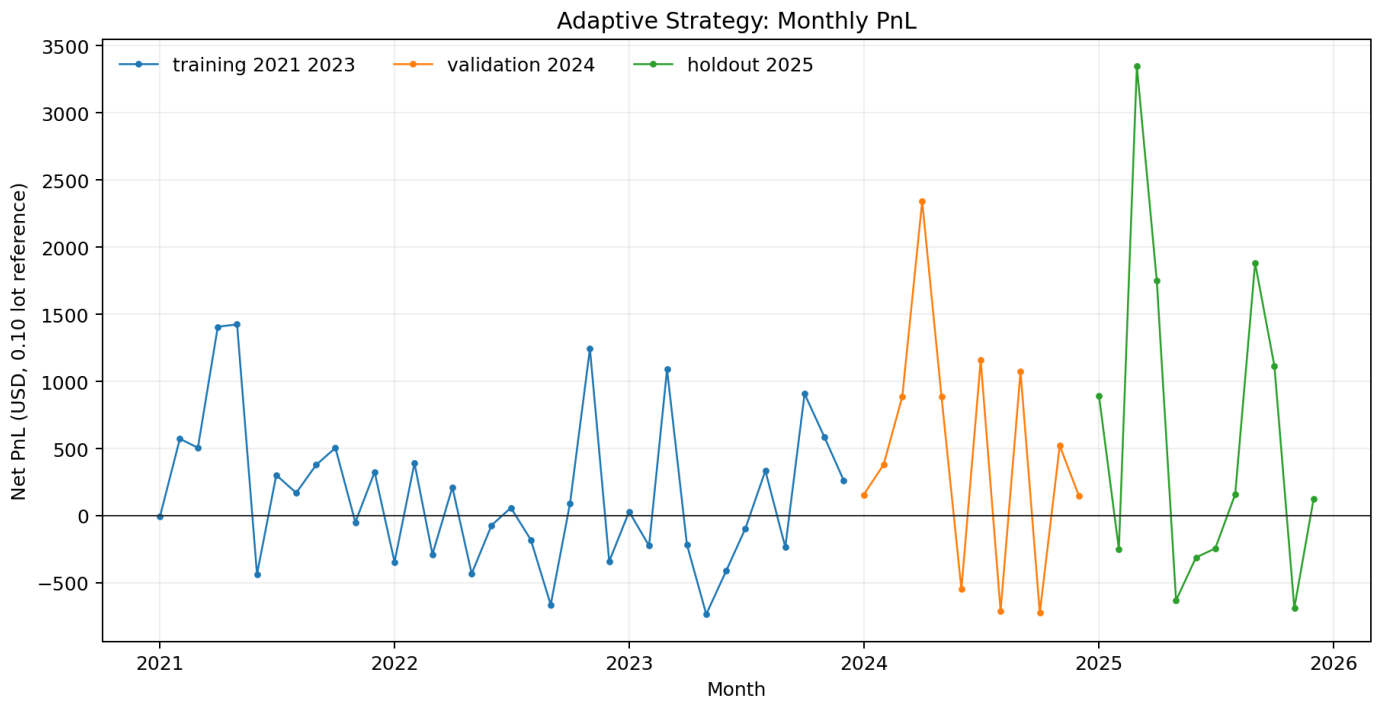


Figure 3: Adaptive monthly PnL

EA input	Default	Purpose
InpEnableNewEntries	false	Requires explicit test activation.
InpFixedLots	0.02	Conservative initial test volume.
InpReturnLookbackBars	3	Three-bar return definition.
InpQuantileBars	5760	20 trading-day M5 threshold window.
InpLowerQuantile	0.20	Oversold return percentile.
InpVolumeMAPeriod	60	Volume normalization window.
InpMinimumVolumeRatio	1.50	Volume surprise threshold.
InpHoldBars	24	120-minute fixed exit.
InpUseHigherTrendFilter	true	Enables completed H1 confirmation.
InpMaxDailyLossPct	2.00	Daily realized-loss lock.
InpMaxConsecutiveLosses	4	Cooldown trigger.
InpCooldownBars	12	Cooldown duration.

The EA uses MT5 tick volume because most XAUUSD brokers do not expose exchange-traded volume. This difference from proxy aggregate-trade volume is one of the main reasons the MT5 test is decisive.

11. MT5 Strategy Tester protocol

Use the broker symbol actually intended for testing, for example **XAUUSD**, **XAUUSD.a**, or the broker's equivalent. Do not assume an arbitrary point value, volume step, contract size, spread, or commission.

Recommended test procedure:

1. Compile the EA in MetaEditor and confirm there are no compiler warnings.
2. Select M5 and the most accurate available real-tick mode.
3. Enable `InpEnableNewEntries=true` only inside Strategy Tester.
4. Run the base configuration with trend filter and locks disabled.
5. Run the adaptive configuration with the documented defaults.
6. Repeat at 0.01, 0.02, and a broker-appropriate risk-sized volume.
7. Include actual commission, swap, and the broker's historical spread.
8. Export the deal list and equity curve for independent calculation.
9. Separate performance by year, session, spread percentile, and volatility.
10. Reject the candidate if positive proxy behavior disappears after realistic broker costs or if drawdown exceeds the predefined account risk budget.

12. Failure modes and rejection criteria

The strategy should be rejected, disabled, or redesigned if any of the following occur in broker-native testing:

- PF falls below 1 after realistic costs over a sufficiently large sample.
- The H1 filter reduces trade count but does not improve risk-adjusted results.
- News, rollover, or spread expansion dominate losses.
- The volume proxy has no useful relationship with broker tick volume.
- A small number of exceptional trades account for most profits.
- The daily loss lock only postpones, rather than reduces, drawdown.
- Results are highly sensitive to one exact EMA, percentile, or hold-period parameter.

The most important rejection criterion is not a single losing month. It is a lack of robustness across independent time periods and realistic broker costs.

13. Reproducibility commands

```
python scripts/research_order_flow_absorption.py \
  data/derived/PAXGUSDT_5m_2021_2025_weekdays.csv \
```

```

data/derived/PAXGUSDt_order_flow_5m_2021_2025.csv \
--factor volume_return_3_reversal --bar-minutes 5 --hold-bars 24 \
--side long --session-hours 0-23 --use-higher-trend-filter \
--max-daily-loss-percent 2 --initial-equity 10000 \
--max-consecutive-losses 4 --cooldown-bars 12 \
--trades outputs/m5_adaptive_volume_reversal_trades.csv \
--report reports/M5_Adaptive_Volume_Reversal_Backtest.md

python scripts/generate_adaptive_factor_paper_charts.py \
  outputs/m5_volume_reversal_holdout_trades.csv \
  outputs/m5_adaptive_volume_reversal_trades.csv \
  --assets reports/assets

```

14. Final conclusion

The evidence supports continued professional testing, not production trading. The base factor has a coherent economic interpretation and positive proxy-data results across its original training, validation, and holdout sequence. The adaptive controls reduce observed recent-sample drawdown but are exploratory because they react to a previously observed drawdown. The next non-negotiable gate is an independent XAUUSD Bid/Ask test with realistic MT5 costs and execution. Until it passes, the EA should remain a test instrument.